

1 Basic Knowledge & SVCDE Methods for ODE

1.1 Linear Algebra || Calculus || SVCDE Method for ODE

Matrix Calculate: $\det(kA) = k^n \det(A)$ || $\det(A^{-1}) = \det(A)^{-1}$ || $(AB)^* = B^* A^*$ || $\det(A) = \det(A^T)$.

- Eigenvalue λ & Eigenvector $\mathbf{x}$: Def: $A\mathbf{x} = \lambda\mathbf{x}$. || Method: $^1 \det(A - \lambda I) = 0 \rightarrow \lambda$. $^2 (A - \lambda I)\mathbf{x} = 0 \rightarrow \mathbf{x}$.
- Arithmetic Manipulate (AM) λ : λ 重复出现的次数. $\geq$ Geometric Manipulate (GM) of λ^T : $\# \mathbf{x}$ under λ .
- Generalised Eigenvectors: If $AM > GM$: Using $\xi_\lambda = (A - \lambda I)\eta_1, \eta_{n-1} = (A - \lambda I)\eta_n$ for each ξ_λ . η_j are generalised eigenvectors.

Some Defs: $\nabla F = (\partial_x F, \partial_y F, \partial_z F, \dots)$ || Simply Connected: 没有洞, 相互连接的区域. || ∂D : 区域 D 的边界.

Method of Integrating Factors: $y' + p(x)y = q(x)$. $\Rightarrow$ Integrating Factor: $\mu = e^{\int p(x)dx}$, $\mu y = \int q(x) \cdot \mu dx$

Homogeneous 2nd ODE form: $ay'' + by' + cy = 0$ Characteristic $ar^2 + br + c = 0$, Roots: r_1, r_2 ps: $y'' + \lambda y = 0$ 见附录

$^1 \Delta > 0, y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$; $^2 \Delta = 0, y = (C_1 + C_2 x)e^{rx}$; $^3 \Delta < 0, y = e^{\alpha x}[C_1 \cos(\beta x) + C_2 \sin(\beta x)]$ ps: $r_1, r_2 = \alpha \pm \beta i$

non-Homogeneous 2nd ODE form: $ay'' + by' + cy = g(x) \Rightarrow y = y_h + y_p$ ps: y_h is sol for $g(x) = 0$.

Method of Undetermined Coefficients: Let $A_n(x) = \sum_{i=0}^n a_i x^i$, $B_n(x) = \sum_{i=0}^n b_i x^i$, $P_n(x) = \sum_{i=0}^n p_i x^i$.

- 1. If $g(x) = P_n(x) \Rightarrow y_p = x^s A_n(x)$, $s := 0, (0 \neq r_1, r_2); 1, (0 = r_1 \text{ or } r_2); 2, (0 = r_1 = r_2)$.
- 2. If $g(x) = M e^{kx} \Rightarrow y_p = C x^s e^{kx}$, $s := 0, (k \neq r_1, r_2); 1, (k = r_1 \text{ or } r_2); 2, (k = r_1 = r_2)$.
- 3. If $g(x) = M \cos(kx) + N \sin(kx) \Rightarrow y_p = x^s (C_1 \cos(kx) + C_2 \sin(kx))$, $s := 0, (r_1, r_2 \neq ik); 1, (r_1, r_2 = ik)$.
- 4. If $g(x)$ = product from 1,2,3. $\Rightarrow y_p$ = corresponding product of y_p (x^s 只乘一次), $s :=$ 令 y_p 与 y_c 没有相同项. (≤ 2)

Euler Equation: General: $a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_0 y = 0 \Rightarrow$ Seek Solution as: $y = x^\alpha \Rightarrow$ solution: $y = \sum_i c_i x^{\alpha_i}$

2.1 Homogeneous

Principle of Superposition: If y_i are solutions $\Rightarrow \sum_i c_i y_i$ also. || $S := \{y | y' = Py\}$ is a linear space, $\dim(S) = n$ if P_{ij} continuous.

Fundamental Set of Solutions (FSoS): If $\{y^{(j)}(t)\}$ linearly independent at $t \in I$, it is FSoS in I .

General Solution: If $\{y^{(j)}(t)\}_{j=1}^n$ is FSoS (Linearly Independent) $\Rightarrow$ General solution: $y = \sum_{j=1}^n c_j y_j$

2.1.1 Eigenvalue/vector Method (A constant)

AM=GM|Real λ : general solution: $y = \sum_j c_j e^{\lambda_j t} \xi_j$.

AM=GM|Complex λ : (ctd.) For $\lambda, \bar{\lambda} \in \mathbb{C} \Rightarrow e^{\lambda t} \xi = \mathbf{u} + v i \Rightarrow$ Independent sol: $\mathbf{u}, \mathbf{v} \Rightarrow$ General solution: $y = c_1 \mathbf{u} + c_2 \mathbf{v} + \dots$

AM>GM;(ctd.) For each ξ_λ , under λ $y^{(1)} = e^{\lambda t} \xi_\lambda$ $y^{(2)} = e^{\lambda t} (t \xi_\lambda + \eta_1)$ $y^{(3)} = e^{\lambda t} (\frac{t^2}{2!} \xi_\lambda + t \eta_2 + \eta_1), \dots$ 类似泰勒展开

2.1.2 Matrix Method | Exponential

Matrix Method: $y(t) = e^{At} y(0)$ (分两种方法: 1. 直接展开; 2. 通过一些已有 property 计算. ps: $e^{At} = \Psi$)

· Property|AM=GM: $e^{At} = T e^{DT} T^{-1}$, $T = [\xi_1, \dots, \xi_n]$, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, $e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$

· Property|AM>GM: $e^{At} = T e^{Jt} T^{-1}$, $T = [\lambda_1, \dots, \lambda_i, \eta_1^{(i)}, \dots, \lambda_j, \eta_1^{(j)}, \dots]$, $J = T^{-1} A T$ (Jordan form), e^{Jt} = 见右边

Diagonalisation Method: (Ctd.) Let $y = T x \Rightarrow x' = J x$ [or $x' = D x$] 按前面方法解

2.1.3 Matrix Method | Fundamental Matrix

Fundamental matrix Ψ : If $\{y^{(j)}(t)\}$ is FSoS. $\Rightarrow \Psi(t) = [y^{(1)}(t), \dots, y^{(n)}(t)]$.

· Property: $\Psi'(t) = P(t)\Psi(t)$ (It's a sol). || $\det[\Psi(t)] = W(t) \neq 0$ || general solution: $y = \Psi c$, $c = \Psi^{-1}(t_0) y_0$

· Relation with e^{At} : $e^{At} = \Psi(t)$ with $\Psi(0) = I$ || $e^{At} = \Psi(t) \Psi^{-1}(0)$

2.2 non-Homogeneous

Theorem: Solution of $(*)$: $y = \sum_i c_i y^{(i)} + y_p$, where $y^{(i)}$ is solution of it's Homogeneous.

⁴ 通常会让特解中积分后的常数 c=0

Variation of Parameters: $y = y_g + y_p$; $y_g = \Psi(t) \Psi^{-1}(t_0) y(t_0)$; $y_p = \Psi(t) u(t)$, $u(t) = \int_{t_0}^t \Psi^{-1}(s) g(s) ds$. 经常 $\Psi(t) u' = g$ 来找 u

Matrix methods|Diagonalisation: 1 Let $y = T x$ $^2 (*) \Rightarrow x' = D x + T^{-1} g = D x + h$ 3 直接计算; $z_i = c_i e^{r_i t} + e^{r_i t} \int_{t_0}^t e^{-r_i s} h_i(s) ds$ $\xrightarrow{y_p}$

Undetermined Coefficients: when $g(t)$ is polynomials, or real/complex exponential. Let $P_n(t) = \sum_{i=0}^n p_i t^i$, $A_n(t) = \sum_{i=0}^n a_i t^i$

1 构造 y_p 与 g 结构相匹配 (e.g. If $g = P_n(t) \Rightarrow y_p = A_n(t)$; If $g = c e^{kt} \Rightarrow y_p = a e^{kt}$) 2 特殊: If $g = c e^{\lambda t}$, λ is eigenvalue with $AM = n \Rightarrow y_p = e^{\lambda t} \sum_{i=0}^n a_i t^i$ 3 带入原方程得未知系数

2.3 Geometrical Interpretation & Linear Approximation

* 此节中*, Consider nonlinear autonomous system: $x' = F(x, y), y' = G(x, y)$ (**)

Stable: Critical Point x_0 is stable iff $\forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall$ sols $x(t)$ with $\|x(0) - x_0\| < \delta$, then $\|x(t) - x_0\| < \epsilon, \forall t \geq 0$ || 直观理解: 开始近 $\rightarrow$ 保持近.

Attracting: Critical point x_0 is attracting iff $\exists \delta > 0$ s.t. $\forall$ sols $x(t)$ with $\|x(0) - x_0\| < \delta$ satisfies: $\lim_{t \rightarrow \infty} x(t) = x_0$ || 直观理解: 开始近 $\rightarrow$ 不断趋向.

Asymptotically Stable: Critical point x_0 is asy.stab iff 1 Stable, 2 Attracting.

Theorem: If $x_*(t) = (x_*, y_*)$ s.t. $F(x_*), G(x_*) \neq 0 \Rightarrow \exists H$ s.t. $H(x, y) = (\bar{x}, \bar{y}), \bar{x}' = 1, \bar{y}' = 0$ H is a smooth change of variable function in neighbourhood of x_* .

Non-Linear $\rightarrow$ Linear: For system (**), assume $x_0 = (x_0, y_0)$ is critical point. $\Rightarrow u = x - x_0$, then $u' \approx Au$ Linear; $A = \begin{pmatrix} \partial_x F(x_0) & \partial_y F(x_0) \\ \partial_x G(x_0) & \partial_y G(x_0) \end{pmatrix}$ jacobian

2.4 Classification of Critical Points

* 此节中*, Consider linear system: $x' = Ax$. Critical Point(s): when $x' = Ax = 0$

Classification of Critical Points at (x_0, y_0) (Linear): r_1, r_2 be sol of $\det(A - \lambda I) = 0$. || $\mathbb{C} : r = \lambda \pm i\mu (\mu > 0)$ || If A constant, write sol: $x = c_1 e^{r_1 t} \xi_1 + c_2 e^{r_2 t} \xi_2$ || $GM = 1: x = c_1 e^{r t} \xi + c_2 e^{r t} (t \xi + \eta)$

$\mathbb{R}/\mathbb{C}$	Condition Stability	Type Name	Phase Plane Description	Other	
$\mathbb{R}$	$r_1 < r_2 < 0$ asy.stab	N NSk	向原点, ξ_2 直线, ξ_1 曲线, 和 ξ_1 周围 $y = \pm x^3$	$c_2 \neq 0, t \rightarrow -\infty, \xi_2$ 主导方向; $c_2 = 0, t \rightarrow -\infty, \xi_1$ 主导方向	PS: N = Node PN = Proper Node IN = Improper or: Degenerate Node SP = Saddle Point SpP = spiral point or: Focus Point C = Center NSk = Nodal Sink NSo = Nodal Source
	$r_1 > r_2 > 0$ unstable	N NSo	原点向外, ξ_2 直线, ξ_1 曲线, 和 ξ_1 周围 $y = \pm x^3$	$c_1 \neq 0, t \rightarrow \infty, \xi_1$ 主导方向; $c_1 = 0, t \rightarrow \infty, \xi_2$ 主导方向	
	$r_1 > 0 > r_2$ unstable	SP SP	$t \rightarrow \infty, \xi_1$ 从原点向外, ξ_2 从外向原点 and: 像 $y = \pm \frac{1}{x}$, 同进同出	$t \rightarrow \pm \infty: x \rightarrow \infty; \quad t \rightarrow -\infty: c_1, c_2 \neq 0, x \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty: c_2 = 0, x \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty: c_1 = 0, x \rightarrow \infty, \xi_2$ 主导	
	$r_1 = r_2 < 0$, GM=2 asy.stab	PN PN or Stable Star	直线 向原点	直线, u_1 / u_2 is t independent	
	$r_1 = r_2 > 0$, GM=2 unstable	PN PN or Unstable Star	直线 从原点向外	直线, u_1 / u_2 is t independent	
	$r_1 = r_2 < 0$, GM=1 asy.stab	IN (AL>Type: SpP) IN (Stable)	S 曲线, 向原点	$t \rightarrow \infty, x \rightarrow 0, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同	
	$r_1 = r_2 > 0$, GM=1 unstable	IN (AL>Type: SpP) IN (Unstable)	S 曲线, 从原点向外	$t \rightarrow \infty, x \rightarrow \infty, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同	
$\mathbb{C}$	$\lambda = 0, \lambda > 0$ unstable	SpP Unstable Focus	向外椭圆 (elliptical) 螺旋	$t \rightarrow \infty, x \rightarrow \infty$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针, 如果 bc<0, 逆时针	
	$\lambda = 0, \lambda < 0$ asy.stab	SpP Stable Focus	向内椭圆 (elliptical) 螺旋	$t \rightarrow \infty, x \rightarrow 0$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针, 如果 bc<0, 逆时针	
	$\lambda = 0$ stable (AL:Indeterminate)	C (AL:C or SpP) C	椭圆 (elliptical) 和 半长轴 ξ 实部方向	Bounded trajectory or $\exists$ Periodic Trajectories	
AL: Almost Linear Sys. For Almost Linear systems: $\mathbf{u}' = \mathbf{A}\mathbf{u} + \boldsymbol{\eta}$, 如果 linear Approximation (去掉 $\boldsymbol{\eta}$), 按照 Linear 考虑. 如果不去 $\boldsymbol{\eta}$, 按照去掉的算 r_1, r_2 , Type 按照正常 / (AL) 的来. (图像会到 (0, 0))					

Some Defs: Separatrix: 将不同区域分开的轨迹; Basin of attraction: 所有趋近 critical point 的轨迹. ps: In any nonlinear ODEs, it's important to determine the basin of attraction.

3 Non-Linear ODE Methods

* 此节中*, Consider two-dimensional autonomous system: $x' = F(x, y), y' = G(x, y)$ and $x = (x, y)^T$ (#) Lyapunov's: 不需要知道微分方程的确切解 critical value. Poincaré-Bendixson: 寻找 limit cycle

Lyapunov Stability Theorem: 1 (#); 2 ICP $x_0 \in D$; 3 $\exists$ 正定且连续 $E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, E(x_0) = 0$; || $^4 \frac{dE}{dt} = \nabla E \cdot x' = \partial_x E \cdot F + \partial_y E \cdot G$ 负 (半) 定 on $D \setminus x_0(D) \Rightarrow x_0$ asp. (stab)

· Lyapunov function: 符合上面 1-4 条件的

Lyapunov Instability Theorem: 1 (#); 2 ICP $x_0 \in D$; $^3 \exists E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, \partial_t E$ 连续, $E(x_0) = 0$; $^4 \nabla$ 领域 $x_0, \exists x_1 E(x_1) > 0$; || $^5 \frac{dE}{dt}$ 正定 on $D \Rightarrow x_0$ unstable

Periodic Solution: Soln x has: $x(t + T) = x(t), \forall x$. || 闭合曲线 $\Leftrightarrow$ Periodic sol || Periodic Solution + 附近有吸引或排斥 $\Rightarrow$ Limit Cycle || when $\dot{r} = 0, \theta = const$. Periodic sol

Limit cycle: Limit cycle $\Rightarrow$ Periodic Solution || Def $t \rightarrow \pm \infty, \exists$ non-closed trajectory asymptotes to it. (inside or outside) ps: It's Orbital Stability

- 1. Limit cycle is 1 asy.stab: $\forall$ trajectories asy. it; 2 semistable: $\exists$ trajectories 单方向进/出; 3 unstable: $\forall$ trajectories away from it.
- 2. Theorem (If $\exists$ limit cycle / closed): $^1 F, G$ 一阶偏导连续 in simply connected D ; $^2 \exists$ 极限环/封闭环 $(*)$ in $D \Rightarrow ^1 \exists$ CP in $(*)$; 2 If $\exists$ CP in $(*)$, it's not SP.
- 3. Theorem (No limit cycle / closed): $^1 F, G$ 一阶偏导连续 in simply connected D ; $^2 \partial_x F + \partial_y G$ 符号不变 on $D \Rightarrow$ No closed trajectory in D .

2 的推论: 1 If simple connected D no CP $\Rightarrow$ No closed trajectory in D . 2 If simple connected D has unique CP $\Rightarrow$ No closed trajectory in D

Poincaré-Bendixson Theorem: $^1 F, G$ 一阶偏导连续 in D ; $^2 \exists D_1 \subset D$ bounded; $^3 R = D_1 \cup \partial D_1$ has NO CP; || $^4 \exists$ soln $x \in R, \forall t \geq t_0 \Rightarrow \Rightarrow ^1 \exists$ closed trajectory. 2 Either x spiral towards a Periodic trajectory. or x periodic $^3 R$ is not simply connected

Find Trapping Region / Limit Cycle: trapping region: 符合上面定理 2-4 除 CP 的 region. Method: polar coordinate, 计算 $r^2 = x^2 + y^2$ 对 t 的求导, 寻找 $rr' \geq 0$ or $rr' \leq 0$

4 Fourier Series

* BVPs 可能没有解/一个解/无数解; IVP 只会会有一个解

4.1 Function Space, Eigenfunctions, Periodic Function and Orthogonal Function

Periodic Functions: Function $f(x)$ is Periodic iff $f(x + T) = f(x), \forall x$. Period: T. Fundamental Period: The smallest possible T. || Property: If f, g has same period $T \Rightarrow af(x) + bg(x)$ also has period T .

Differential Operators: Any linear combination of n -th derivatives defines a linear map or linear operator. e.g. $p(x) \frac{d}{dx} + \frac{d^2}{dx^2} : y \mapsto p(x) \frac{dy}{dx} + \frac{d^2 y}{dx^2}$

Eigenvalues & Eigenfunctions: If D is a differential operator $\Rightarrow$ If $Df = \lambda f$, then λ is Eigenvalue and f is Eigenfunctions || Property: 对于微分方程: $Df = \lambda f \Rightarrow$ solution: $f = \sum c_n f_n$ f_n are Eigenfunctions

Inner Product: $(f, g) = \int_a^b f(x) \overline{g(x)} dx$ For funcs f, g in (a, b) || Orthogonal: $(f, g) = 0$ || Square Norm: (f, f) . || Kronecker delta: $\delta_{mn} = \{0, m \neq n; 1, m = n$.

1.2 Linear || (ODE) Uniqueness Theorem || Autonomous

Def of Linear: Let L 去掉 $y_i^{(n)}$ 的函数 (e.g. $L = \frac{d^2}{dx^2} - x$ for $y'' - xy = x^2$)
 $\Rightarrow$ Linear if: $L[a_1 y_1 + a_2 y_2] = a_1 L[y_1] + a_2 L[y_2]$

Existence & Uniqueness Theorem (Real func):

$^1 y'(t) = F(t, y^T), y(t_0) = y_0$. Let $R_t = [t_0 - \delta, t_0 + \delta] \times \prod_j [y_{0j} - \delta_j, y_{0j} + \delta_j]$.

$^2 F_{ij}, \partial y_i F_{ij} : R_t \rightarrow \mathbb{R}$ are continuous. $\Rightarrow \exists$ unique solution y in R_t'
 $R_t' = [t_0 - \epsilon, t_0 + \epsilon] \times \prod_j [y_{0j} - \epsilon_j, y_{0j} + \epsilon_j] \subseteq R_t$.

Autonomous: For $y'(t) = f(t, y^T)$. $\partial_x f = 0 \Rightarrow$ autonomous. or: $f(t, y^T) = f(y^T)$

2 Linear ODEs

n -th ODE $\rightarrow$ 1st ODEs: For $y^{(n)} = F(t, y, y', \dots, y^{(n-1)})$. $\Rightarrow$ Let $x_i = y^{(i-1)}$.

Def of Linear 1st ODEs: $y'(t) = P(t)y(t) + g(t) (*)$, P is matrix-valued function.

Homogeneous: $g(t) = 0$.

Existence & Uniqueness Theorem: If P_{ij}, g of $(*)$ continuous in $R, y(t_0) = y_0$

$\Rightarrow \exists$ unique solution y in R' $R' = [t_0 - \epsilon, t_0 + \epsilon] \subseteq R = [t_0 - \delta, t_0 + \delta]$

Solve: $y' = Py$ || If P constant, $P = A$ || λ, ξ, η be 特征值/向量/广义特征向量.

Abel's Theorem: If $W(t_0) \neq 0, t_0 \in I \Rightarrow \forall t \in I, W(t) \neq 0$

Abel's Formula: $W(t) = W(t_0) e^{\int_{t_0}^t \text{tr}[P(s)] ds}$; $t_0, t \in I$ $P(t)$ is continuous in I .

Wronskian at x : $W[y^{(1)}, \dots, y^{(n)}](t) = \begin{vmatrix} y_1^{(1)}(t) & \dots & y_1^{(n)}(t) \\ \dots & \dots & \dots \\ y_n^{(1)}(t) & \dots & y_n^{(n)}(t) \end{vmatrix}$.

$\cdot W(t) \neq 0, \forall t \Rightarrow \{y^{(j)}\}$ linearly independent. ps: 经常用 Abel Theorem

$$e^{Jt} = e^{\lambda_i t} \begin{pmatrix} 1 & t & \frac{t^2}{2} & \dots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \dots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

Exponential of Matrix: $e^{At} := \sum_{n=0}^{\infty} \frac{(At)^n}{n!} = I + At + \frac{1}{2!} A^2 t^2 + \dots$
 $:= \lim_{n \rightarrow \infty} (I + \frac{At}{n})^n$

· Property: $e^{A+B} = e^A e^B$ if $AB = BA$ || $e^{A \cdot 0} = I$ || $[e^{At}]' = A e^{At}$

Phase Plane: O_{xy} referred as phase plane.

Trajectories: Sol of ODE. $x(t) = (x(t), y(t))^T$

Phases Portrait: 一组有代表性的轨迹

Critical Point (Equilibrium Point):

$[x' = F(x_0) = 0, y' = G(x_0) = 0]$ or $[x' = 0]$

(critical point 外变化简单)

4.2 Fourier Series

Fourier Series: Periodic function with $T = 2L$, $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}) \right)$ (*)

· **Euler-Fourier Formulas:** Suppose (*) converges in the interval $[-L, L]$. Then: $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx$ $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(\frac{n\pi x}{L}) dx$ $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$ (Average Value of f over $2L$)

Piecewise Continuous: $f(x)$ piecewise continuous on $[a, b]$ iff $\exists a = x_0 < \dots < x_n = b$ s.t. $^1 f(x)$ continuous $\forall x \in (x_{i-1}, x_i)$ $^2 f(x_i^{\pm}) = \lim_{x \rightarrow x_i^{\pm}} f(x)$ finite.

Fourier Convergence Theorem: If $^1 f, f'$ piecewise continuous on $[-L, L]$, $^2 f(x + 2L) = f(x)$. $\Rightarrow$ Fourier Series of f converges. **Fourier 和 SL 都不一定逐点收敛, 但一定均方收敛.** (类似泰勒展开)

· Furthermore, Let Fourier Series of f is F . $F(x) \rightarrow f(x)$, $f(x)$ continuous; $F(x) \rightarrow \frac{f(x^+) + f(x^-)}{2}$, $f(x)$ discontinuous.

Term-By-Term Integration: Fourier Series for *piecewise continuous* functions can be integrated term-by-term to Convergent Fourier Series.

Finite Truncation (Partial Sum): Finite Truncation of Fourier Series is: $s_k = \frac{a_0}{2} + \sum_{n=1}^k \left(a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}) \right)$ || **Error:** $e_N = s_N - f(x)$

· $n \rightarrow \infty$: 1 Continuous points, Fourier approximation of $f(x)$ is better ($|e_n|$ decrease). 2 Discontinuous points, oscillation of Fourier series get smaller.

· **Gibbs Phenomenon:** The Amplitude of the *largest* oscillation (near discontinuous points) does *not decrease*.

4.3 Other Results on Fourier Series

Extension of a function: $f(x)$ know for $x \in [0, L] \Rightarrow$ Even Ext.: $h(x) = \{f(x), [0, L]; f(-x), (-L, 0) \}$ || Odd Ext.: $h(x) = \{f(x), (0, L); 0, x = \pm L, 0; -f(-x), (-L, 0)$

Even Ext: $a_n = \frac{2}{L} \int_0^L f(x) \cos(\frac{n\pi x}{L}) dx, b_n = 0$ || Odd Ext: $b_n = \frac{2}{L} \int_0^L f(x) \sin(\frac{n\pi x}{L}) dx, a_n = 0$

Complex Form of Fourier Series: $f(x) = \sum_{-\infty}^{\infty} c_n e^{\frac{i n \pi x}{L}}$ || $c_n = \frac{a_n - i b_n}{2}, c_{-n} = \frac{a_n + i b_n}{2}; c_0 = \frac{a_0}{2}, n > 0$ || $a_0 = 2c_0; a_n = c_n + c_{-n}; b_n = i(c_n - c_{-n})$

· **Euler-Fourier Formula (Complex):** If (f) of Periodic $f, (T = 2L)$ converges in $[-L, L] \Rightarrow c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-\frac{i n \pi x}{L}} dx$ || $c_{-n} = \overline{c_n}$ if f real.

Parseval's Theorem: If 1 Periodic function f ; $^2 f$ has convergent Fourier Series. $\Rightarrow$ Square Norm $\langle f, f \rangle = \int_{-L}^L |f(x)|^2 dx = 2L \sum_{n=-\infty}^{\infty} |c_n|^2 = L \left[\frac{|a_0|^2}{2} + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2) \right]$ ps: 无限维的勾股定理

5 Linear PDEs by Separation of Variables

Requirement: Assume $u(x, t) = X(x)T(t)!$ $\Rightarrow$ If PDE 能写成类似 $\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} \Rightarrow \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = \lambda$ (Separation Constant) $\Rightarrow$ 可以使用此方法!

1.Homo, 假设: $u(0, t) = u(L, t) = 0; u(x, 0) = f(x)$ 2.Inhomo, 假设: $u(0, t) = a, u(L, t) = b; u(x, 0) = f(x)$

Basic Idea|Homo: 1. 假设解 u 能写成 $u(x, t) = X(x)T(t) \Rightarrow$ 2. 代入 PDE, 得两个 ODE $\Rightarrow$ 3. 解其中一个得到特征函数 $X_n \Rightarrow$ 4. 代 initial 条件得限制条件 $\Rightarrow$ 5. 解另一个得到特征函数 $T_n \Rightarrow$ 6. 特征函数 $X_n T_n$ linear combination 得到 u . 7. 代 boundary 条件用 Euler-Fourier 得到的系数.

· Remark: 1 化简中可以使用正比简化计算 2 对 n 的取值可为 $1, 2, 3, \dots$ 3 对 λ 的取值会出现 $\sin, \cos$ 时为正不容易出错, 或注意正负号

Basic Idea|Inhomo: 1. Let $u(x, t) = v(x) + \omega(x, t), v(x) = \frac{b-a}{L}x + a \Rightarrow \omega(x, t)$ satisfy: $\omega(x, 0) = f(x) - v(x); \omega(0, t) = \omega(L, t) = 0$ 2. 按 HOMO 解 $\omega(x, t)$

Laplace Equation: 2D: $\nabla^2 u = \partial_x^2 u + \partial_y^2 u = 0$ 3D: $\nabla^2 u = \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$ Following are the method to solve:

1. **Rectangle:** $\{0 \leq x \leq a, 0 \leq y \leq b\}$ with $u(x, 0) = u(x, b) = u(0, y) = 0; u(a, y) = f(y)$ 直接按照 Homo 正常计算 $\Downarrow$ 可能用到 Euler Equation

2. **Circle:** $\{u(r = a, \theta) = f(\theta); u(r, \theta)$ bounded $\} \partial_x^2 u + \partial_y^2 u = 0; y = r \cos(\theta), y = r \sin(\theta)$ 变极坐标 $R(r), \Theta(\theta)$ 2 一样的计算 + 条件 $\Theta(\theta + 2\pi) = \Theta(\theta)$

Heat eqn: $\frac{\partial}{\partial t} u = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ - Parabolic

Wave eqn: $\frac{\partial^2}{\partial t^2} u = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ - Hyperbolic

Laplace eqn: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ - Elliptical

一般来说都会 **bounded** 的条件加以限制, 即如果存在 $r \rightarrow 0$ 或 $r \rightarrow \infty$ 会 **unbounded** 那么需要让其系数等于 0

6 Sturm-Liouville Theory (Another 'Fourier')

6.1 Sturm-Liouville Theory

Sturm-Liouville (SL) Problem: Form: $-(p(x)y')' + q(x)y = \lambda r(x)y$, $p, r > 0$ || With: $a_1 y(a) + a_2 y'(a) = b_1 y(b) + b_2 y'(b) = 0$ p, p', q, r continuous. || **weight function:** $r(x)$. || $\Downarrow$ **Lagrange's Identity**

$\Rightarrow$ Let *Linear Operator:* $L(y) := -(p(x)y')' + q(x)y \Rightarrow$ Problem becomes: $L(y) = \lambda r(x)y$ (It's a Eigenvalue Problem $\mathcal{L}(\Psi_n) = \lambda_n r(x)\Psi_n$) $\Rightarrow \forall$ solutions u, v with (二阶导连续) of SL Problem: $\langle L(u), v \rangle = \langle u, L(v) \rangle$!!!

· **Properties|Real|Orthogonal:** In SL Problem: $^1 \lambda_n, \Psi_n$ are *real*; || 2 Eigenfunctions Ψ_1, Ψ_2 with $\lambda_1 \neq \lambda_2$. **Modified Inner Product:** $\langle \Psi_1, \Psi_2 \rangle = \int_a^b r(x) \Psi_1 \Psi_2 dx = 0$ i.e. Ψ_1, Ψ_2 orthogonal.

· **Properties|Simple|Ordered:** In SL Problem: $^3 \lambda_n$ are *simple*: $\forall \lambda_n$ has *unique* Ψ_n ; || $^4 \lambda_n$ are *Ordered*: $\lambda_1 < \lambda_2 < \dots < \lambda_n < \dots \rightarrow \infty$ as $n \rightarrow \infty$

Normalisation of Eigenfunction: $^1 \Psi_n$ normalised if: $\langle \Psi_n, \Psi_n \rangle = 1$ || 2 Method (Normalised Ψ_n): $\Phi_n = \frac{1}{\sqrt{\langle \Psi_n, \Psi_n \rangle}} \Psi_n = [\int_a^b r(x) \Psi_n^2 dx]^{-1/2} \Psi_n$

Sturm-Liouville Convergence Theorem: If f, f' are piecewise continuous on $[0, 1] \Rightarrow F(x) = \sum_{n=1}^{\infty} c_n \Phi_n$ converges. || Φ_n are *orthonormal* eigenfunction of a Sturm-Liouville problem.

· Furthermore, $^1 F \rightarrow f$ when f is continuous; $^2 F \rightarrow \frac{f(x^+) + f(x^-)}{2}$ when f is discontinuous. 边界点右边界条件决定 **Calculate Coefficients:** $c_n = \langle f, \Phi_n \rangle = \int_a^b r(x) f(x) \Phi_n(x) dx$ || ps: c_n minimize the MSE ε_N

· 如果我们不需要归一化: 特征函数 $\Psi_n \Rightarrow f(x) = \sum_{n=1}^{\infty} c_n \Psi_n, c_n = \langle f, \Psi_n \rangle = \int_a^b f(x) \Psi_n(x) dx$ || **Parseval's Identity:** f, f' are piecewise continuous on $[a, b] \Rightarrow \langle f, f \rangle = \int_a^b f^2(x) r(x) dx = \sum_{n=1}^{\infty} c_n^2$

6.2 General Boundary eigenvalue problem || Non-Homogeneous Boundary Value Problems || Open Interval Situation

General Boundary eigenvalue problem: $L(y) = \lambda r(x)y, a < x < b$ || With $n - th$ order operator: $L(y) = p_n(x) \frac{d^2 y}{dx^2} + \dots + p_1(x) \frac{dy}{dx} + p_0(x)y$ || Assume n linear Homogeneous Boundary conditions at the endpoints.

$\Rightarrow$ And suppose that $\langle L(u), v \rangle = \langle u, L(v) \rangle \Leftarrow$ (Self-Adjoint Operator) $\rightarrow L$ must be even order. || **Properties:** 1 Infinite eigenvalues. 2 Orthogonal eigenfunctions. 3 类似 Fourier 可以将任意函数用级数展开.

Non-Homogeneous BVP|Sturm-Liouville: $L(y) = \lambda r(x)y + f(x)$ where $^1 L$ is Sturm-Liouville operator. (就是符合前面 SL Problem operator 的定义 + conditions) $^2 \mu$ is constant.

$\Rightarrow$ Let λ_n, Φ_n be eigenvalues & orthonormal eigenfunctions of $L(y) = \lambda r(x)y$. || Method: **Green's Function:** $G(x, z) = \sum_{n=1}^{\infty} \frac{\Phi_n(x) \Phi_n(z)}{\lambda_n - \mu}$ ($\mu \neq \lambda_n$) $\Rightarrow$ Solution: $y(x) = \int_a^b G(x, z) f(z) dz$ ($\mu \neq \lambda_n$)

· (Ctd.) 1 If $\mu \neq \lambda_n \Rightarrow$ unique solution. 2 If $\exists m, \mu = \lambda_m$: Let $c_m = \int_a^b f(x) \Phi_m(x) dx \Rightarrow$ Case I ($c_m = 0$): Let $b_n = \frac{c_n}{\lambda_n - \mu}, n \neq m$ $y(x) = \sum_{n=1, n \neq m}^{\infty} [b_n \Phi_n(x)] + b_m \Phi_m(x), b_m \in \mathbb{R}$ Case II ($c_m \neq 0$): No Solution.

Signlar / Regular SL Problem: **Signlar:** boundary condition at *closed interval*. || **Regular:** boundary condition at *open interval* or $[a, b]$ or $[a, b]$.

Signlar SL Problem|One-side: $L(y) = \lambda r(x)y$, 奇异点 $x = a, b \Rightarrow \lim_{x \rightarrow a, b} p(\varepsilon) [u'(\varepsilon) v(\varepsilon) - u(\varepsilon) v'(\varepsilon)] = 0 \Rightarrow \forall$ sols $u, v: \int_a^b L(u) v - u L(v) = 0 \Leftarrow \lim_{\varepsilon \rightarrow a, b} p(\varepsilon) = 0$ and u, u', v, v' bounded as $x \rightarrow a, b$

· $\star$ holds $\Rightarrow$ 1 特征值实数 2 正交特征函数 3 可以展开任意函数 || 解题一般思路: 1 分离变量法得到 ODE 2 解 ODE (观察是否是 Bessel 问题), 系数先不管 3 使用边界条件 (注意 bounded) 去除部分无关系数 4 通解: 特征函数线性组合 + series 方法确定系数

6.3 Other Results on Sturm-Liouville Theory

Mean Square Error: $\varepsilon_N = \int_a^b r(x) |f(x) - S_N(x)|^2 dx$ (Truncation) $S_N(x) = \sum_{n=1}^N c_n \Phi_n(x)$ || If $\lim_{N \rightarrow \infty} \varepsilon_N = 0 \Rightarrow F = \sum_{n=1}^{\infty} c_n \Phi_n$ **converge in the mean** to f . (Not Pointwise Convergence)

Complete of Eigenfunctions: $\{\Phi_n\}$ (or $\{\Psi_n\}$) is complete with respect to [Mean Square Convergence] / [Pointwise Convergence] for a *class* of functions $\mathcal{F}$ if $[\lim_{N \rightarrow \infty} \varepsilon_N = 0] / [\lim_{N \rightarrow \infty} S_N(x) = f(x)]$ for all $f \in \mathcal{F}$

Class of Square Integrable functions: $\mathcal{F} = \{f(x) | f \text{ and } f^2 \text{ are integrable on } x \in [a, b]\}$

Theorem|Completeness: The Set orthonormal eigenfunctions $\{\Phi_n\}$ of a *regular* SL boundary problem are *complete* with respect to [Mean Square Convergence]. for a *class* of Square Integrable functions.

7 Laplace's Transform 这里的 L 和前面无关

Laplace Transform: The Laplace transform of f is: $F(s) = L\{f(t)\}(s) = \int_0^{\infty} e^{-st} f(t) dt$ || ps: May not exists

$f \in E$ **xponential Order:** $^1 f$ is piecewise continuous on $[0, \infty)$; $^2 \forall t \geq 0: \exists A, B > 0$ s.t. $|f(t)| \leq A e^{Bt}$

Theorem|Laplace Existence: If $f \in E \Rightarrow \forall s$ sufficiently large, $L\{f(s)\}$ exists.

Theorem|Linear Operator: $L\{c_1 f + c_2 g\}(s) = c_1 L\{f\}(s) + c_2 L\{g\}(s)$ $L(0) = 0$

Theorem|Derivative: If f continuous on $[0, \infty)$; $f, f' \in E \Rightarrow L\{f'\}(s) = s L\{f\}(s) - f(0)$

$\cdot f, f', \dots, f^{(n-1)} \in E$ and continuous $\Rightarrow L\{f^{(n)}\} = s^n L\{f\}(s) - s^{n-1} f(0) - \dots - s f^{(n-2)}(0) - s^0 f^{(n-1)}(0)$

Inverse Laplace Transform: $f(t) = L^{-1}\{F(s)\}(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} F(s) ds, \Re(s) = \gamma$ (general form)

$f(t)$	$L\{f\}(s)$	$f(t)$	$L\{f\}(s)$	$f(t)$	$L\{f\}(s)$
1	$\frac{1}{s}$	$t^n e^{at}$	$\frac{n!}{s^{n+1}} \frac{1}{s-a}, x > a$	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$
$\Theta \sinh(at)$	$\frac{a}{s^2 - a^2}$	$\sin(at)$	$\frac{a}{s^2 + a^2}$	$\cos(at)$	$\frac{s}{s^2 + a^2}$
$\Theta \cosh(at)$	$\frac{s}{s^2 - a^2}$	$e^{at} \sin(bt)$	$\frac{b}{(s-a)^2 + b^2}$	$e^{at} \cos(bt)$	$\frac{(s-a)^2 + b^2}{(s-a)^2 + b^2}$
$u_c(t)$	$\frac{e^{-sc}}{s}$	$t^n e^{at} \sin(bt)$	$(-1)^n \frac{b^n}{ds^n} \frac{1}{(s-a)^2 + b^2}$	$t^n e^{at} \cos(bt)$	$(-1)^n \frac{d^n}{ds^n} (\frac{s-a}{(s-a)^2 + b^2})$

ps: 一般: $s > 0$ || $\sinh, \cosh: s > |a|$ || $e^{at} \sin/\cos(t^n): s > a$ || **• Heaviside Unit Function:** $u_c = X(c, \infty)$

Uniqueness & Existence: 1 If f, g continuous $\Rightarrow L\{f\} = L\{g\} \Leftrightarrow f = g$

2 If f, g pointwise continuous $\Rightarrow$ 唯一性不一定成立 (非连续点上不同) (No practical significance in applications)

8 Appendix

Wronskian Geometric Meaning: Geometrically, W is the volume spanned by the three vectors $y^{(1)}, y^{(2)}, y^{(3)}$; t : 增加或减小代表在 $y^{(i)}$ 方向上 compress/stretch.

和差化积/积化和差: $s \pm s = 2cs, s - s = 2cs, c + c = 2cc, c - c = -2s$ || e.g. $\sin A + \sin B = 2 \sin(\frac{A+B}{2}) \cos(\frac{A-B}{2})$ $\sin A \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$ || $\sin(2x) = 2 \sin(x) \cos(x)$ $\cos(2x) = 2 \cos^2(x) - 1$ $\cos(x) = \sqrt{\frac{1 + \cos(2x)}{2}}$ $\sin(x) = \sqrt{\frac{1 - \cos(2x)}{2}}$

Hyperbolic Func: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$ $\cosh^2(x) - \sinh^2(x) = 1$ $[\sinh(x)]' = \cosh(x)$ $[\cosh(x)]' = \sinh(x)$ || $\sinh(x \pm y) = \sinh(x) \cosh(y) \pm \cosh(x) \sinh(y)$ $\cosh(x \pm y) = \cosh(x) \cosh(y) \pm \sinh(x) \sinh(y)$

$\cdot \sinh(x)$ 单调递增, odd, iff $x = 0, \sinh(x) = 0$ $\cosh(x) > 0$ 单调递增, even || $\sinh(2x) = 2 \sinh(x) \cosh(x)$ $\cosh(2x) = \cosh^2(x) + \sinh^2(x)$

Common Int: $\int \tan(x) dx = -\ln|\cos(x)|$; $\int \cot(x) dx = \ln|\sin(x)|$; $\int \sec(x) dx = \ln|\sec(x) + \tan(x)|$; $\int \csc(x) dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

Common Der: $[\tan(x)]' = \sec^2(x)$; $[\cot(x)]' = -\csc^2(x)$; $[\sec(x)]' = \sec(x) \tan(x)$; $[\csc(x)]' = -\csc(x) \cot(x)$; $[\arctan(x)]' = \frac{1}{1+x^2}$ || $1 + \tan^2(x) = \sec^2(x)$; $1 - \csc^2(x) = \cot^2(x)$; $\csc^2(x) - \cot^2(x) = 1$.

Polar Coordinate: $x = r \cos(\theta), y = r \sin(\theta)$ we have: $r \dot{r} = x \dot{x} + y \dot{y}$, $y \dot{x} - x \dot{y} = -r^2 \dot{\theta}$

Explain of Fourier Series: In application, it's decomposition of signal into infinite (countable) waves; each wave $\cos(\frac{n\pi x}{L}), \sin(\frac{m\pi x}{L})$ has discrete period; a_n, b_m correspond to amplitudes of waves. || In vector algebra, it's a space of periodic functions; f corresponds to an arbitrary vector v ; orthogonal basis $\{1, \cos(\frac{n\pi x}{L}), \sin(\frac{m\pi x}{L})\}$; a_n, b_m correspond to coefficient of each components of basis.

Even & Odd Function: Even: $f(-x) = f(x)$; Odd: $f(-x) = -f(x)$. || Property: $^1 f(x)$ odd: $\int_{-L}^L f(x) dx = 0$ $^2 f(x)$ even: $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$ || 1 eve + eve = eve; 2 odd + odd = odd; 3 eve x eve = eve; 4 odd x odd = eve; 5 eve x odd = odd.

Euler Formula: $e^{ix} = \cos(x) + i \sin(x)$ || $e^{inx} = \cos(nx) + i \sin(nx)$ || $\cos(nx) = \frac{e^{inx} + e^{-inx}}{2}$ || $\sin(nx) = \frac{e^{inx} - e^{-inx}}{2i}$

Solve $y'' + \lambda y = 0$: $^1 \lambda < 0: y = c_1 \sinh(\sqrt{-\lambda} x) + c_2 \cosh(\sqrt{-\lambda} x)$ $^2 \lambda = 0: y = c_1 x + c_2$ $^3 \lambda > 0 (c): y = c_1 \cos(\sqrt{\lambda} x) + c_2 \sin(\sqrt{\lambda} x)$ || **Solve $y' + \beta y = 0$:** $y = c_1 e^{-\beta x}$

Approximation $\sin(x) \approx x$ when $x \rightarrow 0$

Bessel's Equation: $y'' + \frac{1}{x} y' + (1 - \frac{m^2}{x^2}) y = 0 \Rightarrow$ Solution: $y(t) = c_1 J_m(t) + c_2 Y_m(t)$ || $J_m(t)$ is Bessel function of first kind; $Y_m(t)$ is Bessel function of second kind. || ps: $Y_m(t)$ not bounded as $t \rightarrow 0, J_m(t)$ bounded.

8.1 Some Applications

Generalised Heat Equation: $r(x) \partial_t u = \partial_x [p \partial_x u] - qu + F \Rightarrow$ 1 考虑 Homo: 分离变量, 得到一个 SL 问题; 一个简单 ODE. ODE 不计算. 2 将 F/r 进行 SL 分解为 Series. 3 考虑解 $u = \sum X_n(x) b_n(t), b_n$ 未知; $2, 3$ 步入原始方程, 根据特征向量正交得系数关系: 一个 ODE. 4 求解 ODE 系数由其他边界条件给出.